SCNC3111 Group Project Plan | Subject area: Physics

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1. Main topic

We will study how superconducting quantum computing moved from observing quantized energy levels in a single Josephson junction (1985) to correcting quantum errors below the surface-code threshold (2025). The 2007 charge-insensitive qubit design links these periods. Our central question is how a circuit showing quantum behavior became a platform that can protect encoded quantum information. The comparison shows evolution from testing quantum mechanics in a macroscopic circuit to engineering many qubits and reducing logical errors.

2. Research articles from different time periods

[1] Martinis, J. M., Devoret, M. H., & Clarke, J. (1985). Energy-Level Quantization in the Zero-Voltage State of a Current-Biased Josephson Junction. *Physical Review Letters*, 55, 1543–1546. https://doi.org/10.1103/PhysRevLett.55.1543

[2] Koch, J., et al. (2007). Charge-insensitive qubit design derived from the Cooper pair box. *Physical Review A*, 76, 042319. https://doi.org/10.1103/PhysRevA.76.042319

[3] Google Quantum AI and Collaborators. (2025). Quantum error correction below the surface code threshold. *Nature*, 638, 920–926. https://doi.org/10.1038/s41586-024-08449-y

3. Brief findings from the comparison

Scientific goals and evidence. The goal changed from establishing energy quantization in a macroscopic circuit to protecting quantum information. In [1], microwave-induced resonances in the escape rate matched predicted level spacings. The experiment used escape-rate statistics and did not yet show coherent qubit operation. In [3], increasing the surface-code distance by two lowered the logical error rate by a factor of 2.14 ± 0.02. The distance-7 memory used 101 physical qubits. Its logical error probability was (0.143 ± 0.003) % per correction cycle.

Circuit control and readout. Control moved from bias-current tuning and escape-rate readout of one junction [1] to entangling gates, repeated parity checks and real-time decoding in a two-dimensional qubit array [3]. This needed the 2007 design [2]. It reduced charge-noise sensitivity and kept the qubit levels separated enough for control.

Noise. Noise remains the main limit. [1] dealt with thermal activation and damping. [2] made the qubit less sensitive to charge noise by using a large shunting capacitance. [3] also has to handle leakage, correlated errors and decoding overhead. The strategy moved from physical isolation, to a noise-insensitive qubit design, to active logical error correction. This links improved single circuits to the use of many circuits for logical protection.